

```
1
2  #include <stdio.h>
3
4  #define MAX 10
5
6  int graph[MAX][MAX];
7  int visited[MAX];
8  int queue[MAX];
9  int front = -1, rear = -1;
10 int vertices;
11
12 // Queue operations for BFS
13 void enqueue(int value)
14 {
15     queue[++rear] = value;
16 }
17
18 int dequeue()
19 {
20     return queue[++front];
21 }
22
23 // BFS Traversal
24 void BFS(int start)
25 {
26     int i;
27
28     printf("BFS Traversal: ");
```

```
30 visited[start] = 1;
31 enqueue(start);
32
33 while(front != rear)
34 {
35     int node = dequeue();
36
37     printf("%d ", node);
38
39     for(i = 0; i < vertices; i++)
40     {
41         if(graph[node][i] == 1 && visited[i] == 0)
42         {
43             visited[i] = 1;
44             enqueue(i);
45         }
46     }
47 }
48
49
50 // DFS Traversal
51 void DFS(int start)
52 {
53     int i;
54
55     printf("%d ", start);
56
57     visited[start] = 1;
```

```
58
59     for(i = 0; i < vertices; i++)
60     {
61         if(graph[start][i] == 1 && visited[i] == 0)
62         {
63             DFS(i);
64         }
65     }
66 }
67
68 int main()
69 {
70     int edges;
71     int i, src, dest;
72     int start;
73
74     printf("Enter number of vertices: ");
75     scanf("%d", &vertices);
76
77     // Initialize graph
78     for(i = 0; i < vertices; i++)
79     {
80         visited[i] = 0;
81
82         for(int j = 0; j < vertices; j++)
83         {
84             graph[i][j] = 0;
85         }
86     }
87 }
```

```
87
88 printf("Enter number of edges: ");
89 scanf("%d", &edges);
90
91 // Input edges
92 for(i = 0; i < edges; i++)
93 {
94     printf("Enter edge (source destination): ");
95     scanf("%d %d", &src, &dest);
96
97     graph[src][dest] = 1;
98     graph[dest][src] = 1; // Undirected graph
99 }
100
101 printf("Enter starting vertex: ");
102 scanf("%d", &start);
103
104 // BFS
105 BFS(start);
106
107 // Reset visited array for DFS
108 for(i = 0; i < vertices; i++)
109 {
110     visited[i] = 0;
111 }
112
113 printf("\nDFS Traversal: ");
114
115 // DFS
```

```
93 {
94     printf("Enter edge (source destination): ");
95     scanf("%d %d", &src, &dest);
96
97     graph[src][dest] = 1;
98     graph[dest][src] = 1; // Undirected graph
99 }
100
101 printf("Enter starting vertex: ");
102 scanf("%d", &start);
103
104 // BFS
105 BFS(start);
106
107 // Reset visited array for DFS
108 for(i = 0; i < vertices; i++)
109 {
110     visited[i] = 0;
111 }
112
113 printf("\nDFS Traversal: ");
114
115 // DFS
116 DFS(start);
117
118 return 0;
119 }
120
121
```

```
4 #define MAX 10
Enter number of vertices: 3
Enter number of edges: 3
Enter edge (source destination): 2
4
Enter edge (source destination): 2
4
Enter edge (source destination): 5
2
Enter starting vertex: 6
BFS Traversal: 6
DFS Traversal: 6
...Program finished with exit code 0
Press ENTER to exit console.
```